15. CAREER PATHWAYS AND THE FUTURE OF LOW-CODE DEVELOPMENT

Aim

To apply performance optimization techniques—field indexing, efficient subform structuring, and Deluge query tuning—in the Insurance Claim Management System to improve search speed, reduce response latency, and optimize database load.

Theory

As the Insurance Claim Management System grows and stores more claim records, inefficient data structures and scripts can cause performance bottlenecks. Zoho Creator provides techniques to optimize searching, data entry, and record processing.

1. Field Indexing

Indexing frequently searched fields can improve lookup and filtering performance. In the insurance claim application, useful candidates may include Claim ID, Policy Number, Claimant Email, Status, or Insurance_Branches, depending on which fields are searchable and which indexing options the Creator interface supports. Use unique constraints only where values are genuinely required to be unique.

2. Efficient Subform Usage

Subforms can group related repeating details within a parent claim record. For example, if a claim contains multiple evidence items or expense entries, a subform may hold each item's description, amount, and supporting document reference. Use subforms where repeated child entries are needed; keep standalone forms and lookups where independent records or relationships are more appropriate.

3. Deluge Query Tuning and Best Practices

Optimize Deluge scripts by using precise criteria, retrieving only the records needed, and avoiding repeated database reads or writes inside loops. For example, filter claims by Status or Insurance_Branches rather than fetching every claim. Use supported pagination/range options when processing large record sets, and validate the exact syntax against the Creator version and query context.

Procedure

Step 1: Open the Application
Open the existing Insurance Claim Management System in the Zoho Creator Developer workspace and identify the forms and reports that contain or search large amounts of claim data.

Step 2: Review Searchable Fields
Identify frequently searched fields such as Claim ID, Policy Number, Status, and branch. Check which field indexing or uniqueness options are available in the form settings and apply them only where appropriate.

Step 3: Structure Repeating Details
If a claim needs multiple expense or evidence entries, consider a subform with fields such as Evidence Type, Description, Amount, and Document Reference. Avoid creating unnecessary subforms if the current application does not require repeating child data.

Step 4: Optimize Deluge Scripts
Open the relevant workflow or custom function. Replace broad record fetches with precise criteria, limit the number of records processed where possible, and move repeated reads/writes outside loops when practical.

Step 5: Test and Compare Performance
Use test data and the available debugger or execution logs to compare search, submission, and report-loading behavior before and after changes. Record observed timings and check that the results remain correct.

Result

The Insurance Claim Management System can use field indexing, suitable subform design, and optimized Deluge queries to improve responsiveness as claim volume grows. Actual performance gains should be confirmed by testing in Zoho Creator; they depend on the data volume, application design, and available platform features.